

Monash University

FIT5202 - Data processing for Big Data 2026 S1

Assignment 2A: Building ML Models for Road Safety Prediction

Due: **(23:55 Friday 22/May/2026 End of Week 11)**

Worth: 15% of the final marks

Background

Road safety is a paramount concern for modern smart cities. By understanding the underlying patterns that contribute to traffic accidents, local authorities can optimise emergency response times, improve road infrastructure, and implement dynamic traffic calming measures.

In a traffic accident, the "Golden Hour" refers to the critical 60-minute window following a traumatic injury, during which prompt medical treatment offers the best chance of survival. In a modern metropolitan transport network, road safety is a paramount concern. However, when an accident first occurs, the initial reports—often triggered by automated vehicle telemetry, roadside cameras, or panicked civilian emergency calls—are chaotic and lack precise details on severity.

Emergency dispatchers, with limited resources, face a critical resource allocation problem: should they send a standard patrol car to assess the situation or dispatch multiple ambulances, fire crews, and heavy rescue vehicles immediately? Sending too much leaves the city vulnerable elsewhere; sending too little costs lives.

Big data processing enables us to analyse vast amounts of data and identify the key drivers of serious accidents. We can use a detailed historical record of road accidents, including environmental conditions, road characteristics, and vehicle manoeuvres, to build models to assist emergency responses.

Objective of the Project

In this assignment, you will act as a data scientist for a national transport authority. In the previous assignment (A1), you used Apache Spark to study historical data; it might have given you some insight into the data set, for example, which feature/attribute contributes to a fatal accident. Then, we will build a model to predict the severity of an accident based on environmental, human, and road conditions (assignment 2A) and later use it to develop a real-time smart-city intelligence dispatch dashboard for the emergency response centre (assignment 2B).

This assignment is a two-part assignment:

In **Assignment 2A**, we will use Apache Spark's MLlib to construct and train machine learning models. Our focus will be on building a high-quality model.

Later, **Assignment 2B** will utilise Apache Spark Structured Streaming and our trained ML model from Assignment 2A to process a live data stream and make dynamic, real-time predictions and visualisation.

The Datasets:

The dataset can be downloaded from Moodle, it was derived from a mixture of datasets from UK's road accidents.

You will find the following files after extracting the zip file:

- a) a2_accidents.csv: Contains casualty data in accidents (collisions)
- b) a2_collision.csv: Collisions case information
- c) a2_vehicle.csv: Vehicle information about the collision
- d) metadata.xlsx: Metadata about different columns in Excel format.
- e) severity_rating.csv: The severity ratings for collisions, on a scale of 1-10.

More information and a brief description of the columns are in the appendix.

The structure of the dataset is similar to A1, with the removal of some rows/columns to reduce your workload in this assignment.

What you need to achieve

Use case	Based on the historical dataset, build an ML model to predict the severity of a traffic accident.	Classification or Regression
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Architecture

The following figure represents the overall architecture of the assignment setup. **Part A** of the assignment consists of preparing the data, performing data exploration and feature extraction, and building and persisting machine learning models.

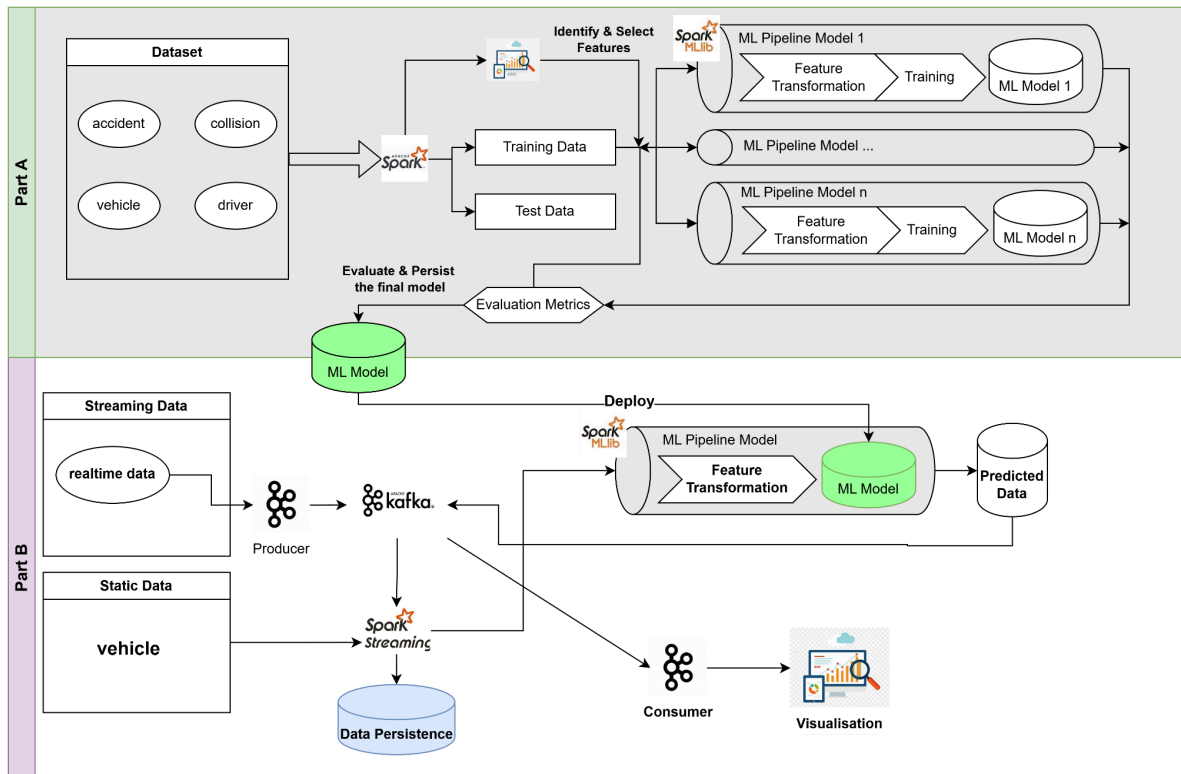


Fig 1: Overall Architecture for Assignment 2 (This assignment is Part A)

In both parts, you must implement the solutions using PySpark DataFrame/MLlib for the data preprocessing and machine learning pipelines. Pandas is only allowed for local plotting and data visualisation. Please follow the steps to document the processes and write the code in your Jupyter Notebook.

Getting Started

- Download the datasets from Moodle.
- Download a template file for submission purposes:
 - **A2A_template.ipynb** file in Jupyter Notebook to write your solution. Rename it into the format (for example, **A2A_xxxx0000.ipynb**. **xxxx0000** is your authcate ID.
- You will use Python 3+ and PySpark 3.5.0+ for this assignment (This environment is the same as we used in labs.)

IMPORTANT:

Please answer each question in your Jupyter notebook file using code/markdown cell. Acknowledge any ideas or codes you referenced from others in the markdown cell or reference list.

If you use generative AI tools, all prompts you use should also be included in the reference section.

You are allowed to modify and reuse your *own* code from A1.

Part 1: Data Loading, Transformation and Exploration (45%)

In this section, you must load the given datasets into **PySpark DataFrames** and use **DataFrame functions** to process the data. For plotting, various visualisation packages can be used, but please ensure that you include instructions to install the additional packages and that the installation will succeed in the provided Docker container (in case your marker needs to clear the notebook and rerun it).

1.1 Data Loading (5%)

1. Write the code to create a SparkSession. For creating the SparkSession, you need to use a SparkConf object to configure the Spark app with a proper application name, to ensure the maximum partition size does not exceed 32MB, and to run locally with all CPU cores on your machine¹ (note: if you have insufficient RAM, reducing the number of cores is acceptable.) (2%)
2. Write code to define the schemas for the datasets, following the data types suggested in the metadata. (2%)
3. Using your schemas, load the CSV files into separate data frames. Print the schemas of all data frames. (1%)

1.2 Data Transformation and Feature Creation (25%)

In this section, we primarily have three tasks:

- 1) **Aggregation (5%)**: The dataset contains multiple rows per accident across different CSV files; group the data by collision_index to create a data frame with the necessary attributes aggregated, based on your understanding of the dataset and subsequent tasks. You will use the "severity_rating" column from the corresponding CSV (on a scale from 1-10) as your label.
- 2) **Imputation (10%)**: The dataset may contain missing values or codes representing "Unknown"; however, in an emergency scenario, it's very common to have missing information as an accident occurs. Some missing data can be filled (imputed); for example, the road surface can be predicted (imputed) if weather conditions are known. Please **explore, discuss** and identify **2 attributes** with missing values that can be imputed. You can use na.fill(), use Spark MLlib's Imputer (or an appropriate imputation strategy), or build your own imputer to handle these missing values based on your understanding of the dataset.

Refer to the Spark MLlib imputation API and fill in the missing values in the dataframe using an appropriate imputer:

<https://spark.apache.org/docs/latest/api/python/reference/api/pyspark.ml.feature.Imputer.html>

¹ More information about Spark configuration can be found in <https://spark.apache.org/docs/latest/configuration.html>

(Note: Although we don't cover imputation in the applied class, students at the master's level are expected to show their ability to study/research in a small task.)

- 3) We know that different times may affect the severity of an accident, based on traffic density. For example, traffic moves slowly during the morning peak time, which reduces the likelihood of severe traffic accidents with many casualties; on the other hand, there are more pedestrians on the road.

Feature Engineering(10%):

- 1) Traffic density varies significantly by time of day. Extract the "Hour" from the accident timestamp and create a new categorical column, Peak_Traffic (e.g., categorising 07:00-09:00 and 16:00-18:00 as "Peak" and the others as "off-peak").
- 2) Please explore, identify and create another feature. This can be based on your understanding of historical data exploration from A1 or domain knowledge from the use case.

1.3 Exploring the data (15%)

1. With the flattened data frame from the previous step, write code to show the basic statistics: a) For each numeric column, show count, mean, stddev, min, max, 25 percentile, 50 percentile, 75 percentile; b) For each non-numeric column, display the top-5 values and the corresponding counts; c) For each boolean column, display the value and count. (note: pandas describe is allowed for this task.) **(5%)**
2. Explore the dataframe and write code to present **two plots of multivariate analysis**. Describe your plots and discuss the findings from the plots. **(10% each)**
 - 150 words max for each plot's description and discussion
 - Feel free to use any plotting libraries: matplotlib, seaborn, plotly, etc. You can refer to <https://samplecode.link>

(Hint 1: You shall only explore the pre-crash variables in our use case to predict severity, as some attributes in the dataset are only possible to collect post-crash.)

(Hint 2: The dataset may be highly skewed, with low-severity accidents as the majority. Think ahead at this stage: How do you handle this for your model performance?)

Part 2. Feature extraction and ML training (35%)

In this section, you must use **PySpark DataFrame functions and ML packages** for data preparation, model building, and evaluation. Other ML packages, such as scikit-learn, would receive **zero** marks.

2.1 Discuss the feature selection and prepare the feature columns (10%)

1. Based on the data exploration from 1.2 and considering the use case, discuss the importance of those features (For example, which features may be noise

and should be removed, which feature has a significant impact on the label column, which should be transformed), which features are you planning to use? Discuss the reasons for selecting them and how you create/transform them²

- 300 words max for the discussion
 - Please only use the provided data for model building
 - You can create/add additional features based on the dataset
 - Hint - Use the insights from the data exploration/domain knowledge/statistical models to consider whether to create more feature columns, or whether to remove some columns
2. Write code to create/transform the columns based on your discussion above

2.2 Preparing Spark ML Transformers/Estimators for features, labels, and models (5%)

Write code to create Transformers/Estimators for transforming/assembling the columns you selected above in 2.1, and create ML model Estimators for Random Forest (RF) and Gradient-boosted tree (GBT) models. Create **two** pipelines.

- **Please DO NOT fit/transform the data yet.**

Please **choose** to create either a multi-class classification model (i.e. 1-10 as 10 categorical classes) or a regression model (i.e. output 1-10 as numerical values). Which one is better? Justify your choice and create your estimators (i.e. RF+GBT for either classification or regression of your choice).

2.3 Training and evaluating models (20%)

1. Write code to split the data for training and testing purposes (70/30), using seed=2026. Use the transformer and estimators in 2.2 to create two ML pipelines. (5%)
2. Based on the model of your choice, discuss the metrics you should use to evaluate the model's performance and the reason you chose them (5%). You are also allowed to create your own custom metrics.
3. Train your model with **both** pipelines³, evaluate the performance for both models (i.e., show the evaluation metrics) and save the **better** model (you need it for Part B of Assignment 2). (10%)

(Note: You may need to go through a few training loops or use more data to create a better-performing model.)

Part 3. Performance Tuning (20%)

Using the better model identified in Part 2, apply hyperparameter tuning techniques to further optimise its performance.

1. Use tools such as ParamGridBuilder and CrossValidator (or TrainValidationSplit) to search for the best combination of hyperparameters for your chosen model.

² This is an open question in which you would need to decide what columns to use as features and what transformation(s) would be required for each feature. Include references when you use arguments from third parties or generative AI tools.

³ Each model training might take from minutes to hours, depending on the complexity of the pipeline model, the amount of training data, the computing power of your laptop and the code efficiencies

2. Document your process, including the parameters you chose to tune, the range of values you tested, and the evaluation metric you used for tuning. Briefly discuss the results and the impact on model performance (no word limit, but keep it concise).
3. Save your tuned model for part B of the assignment (if the model is better than the one you trained in part 2).

(Note: This part of the assignment is marked based on your process and quality of the discussion/understanding.)

Submission

You should submit your final version of the assignment solution online via Moodle. You must submit the files created:

- Your jupyter notebook file A2A_xxxx0000.ipynb
- **A pdf file** saved from jupyter notebook with all output following the file naming format as follows: **A2A_xxxx0000.pdf**

Note that both submitted (ipynb and pdf) files will be scanned using plagiarism detection software. The highest similarity score among students may be interviewed to prove the originality of the task.

Assignment Marking Rubric

Detailed mark allocation is available in each task. For complex tasks and explanation questions, you will receive marks based on the quality of your work.

In your submission, the jupyter notebook file should contain the **code and its output**. It should follow *programming standards, readability of the code, and organisation of code*. Please find the PEP 8 -- Style Guide for Python Code for your reference. Here is the link: <https://peps.python.org/pep-0008/> Penalty applies if your code is hard to understand with insufficient comments.

Other Information

Where to get help

You can ask questions about the assignment in the Assignments section in the Ed Forum, which is accessible on the unit's Moodle Forum page. This is the preferred venue for assignment clarification-type questions. **You should check this forum regularly, as the responses of the teaching staff are "official" and can constitute amendments or additions to the assignment specification.** Also, you can attend scheduled consultation sessions if the problem and the confusion are still unresolved.

Searching and learning on commercial websites/forums (e.g. Quora, Stack Overflow) is allowed. However, you should not post/ask assignment questions on those forums.

Plagiarism and collusion

Plagiarism and collusion are serious academic offences at Monash University. Students must not share their work with any other students. Students should consult the policy linked below for more information.

<https://www.monash.edu/students/academic/policies/academic-integrity> See also the video linked on the Moodle page under the Assignment block.

Students involved in collusion or plagiarism will be subject to disciplinary penalties, which can include:

- The work not being assessed
- A zero grade for the unit
- Suspension from the University
- Exclusion from the University

Late submissions and Special Consideration

ALL Special Consideration, including within the semester, is now handled centrally. This means that students **MUST** submit an online Special Consideration form via Monash Connect. For more details, please refer to the **Unit Information** section in Moodle.

There is a **5% penalty per day including weekends** for a late submission. Also, the cut-off date is 7 days after the due date. No submission will be accepted (i.e. zero mark) after the cut-off date unless you have a special consideration.

Mark Release and Review

- Mark will be released within 10 business days after the submission deadline.
- Reviews and disputes regarding the mark will be accepted a maximum of 7 days after the release date (including weekends).

Generative AI Statement

As per the University's [policy](#) on the guidelines and practices pertaining to the usage of Generative AI:

AI & Generative AI tools may be used SELECTIVELY within this assessment.

Where used, AI must be used responsibly, clearly documented and appropriately acknowledged (see Learn HQ).

Any work submitted for a mark must:

1. Represent a sincere demonstration of your human efforts, skills and subject knowledge that you will be accountable for.
2. Adhere to the guidelines for AI use set for the assessment task.
3. Reflect the University's commitment to academic integrity and ethical behaviour.

Inappropriate AI use and/or AI use without acknowledgement will be considered a breach of academic integrity.

The teaching team encourage students to apply their own critical thinking and reasoning skills when working on the assessments with assistance from GenAI. Generative AI tools may produce inaccurate content, and this could have a negative impact on students' comprehension of big data topics.

Data source acknowledgement:

The dataset is a combination of several real-world and synthetic datasets. We thank the original contributors of the datasets.